Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 2 Quiz 1 - Kinematics in Two Dimensions

Mock Exam B · Form 1: Single-step application

Multiple Choice Questions: Select the best answer. (1 point each)

1. A projectile passes its apex with positive horizontal speed and negligible drag. What angle separates velocity and acceleration at that instant?

(A) 45°

(B) 0°

(C) 90°

(D) 180°

2. A ball is launched horizontally at 4 m/s from a table 4.9 m high. Ignore drag. Find horizontal range. Use g = 9.8 m/s².

(A) 2 m

(B) 4 m

(C) -4 m

(D) 8 m

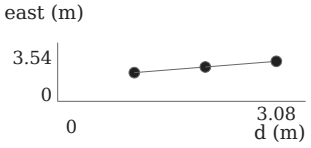
3. A map route starts at east coordinate 2 m and runs at 60° north of east. d (m) -> east (m): (1.08, 2.54); (2.08, 3.04); (3.08, 3.54). What signed vertical change follows a decrease of 0.5 horizontally?

(A) 0

(B) -0.125

(C) -0.25

(D) -0.375



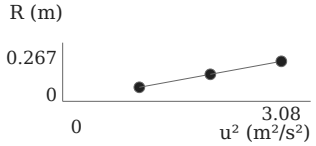
4. Launch at 30° and land at launch height, without drag. Use g=10 m/s². u² (m²/s²) -> R (m): (1.08, 0.09353); (2.08, 0.1801); (3.08, 0.2667). Predict the reading at horizontal coordinate 2.58.

(A) 0.3128

(B) 0.1341

(C) 0.2234

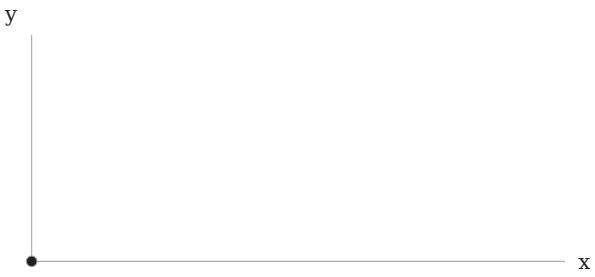
(D) 0.4022



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A ball is launched from ground at 15 m/s, 25° above horizontal, and lands on the same ground. Ignore drag.

a. Draw the initial velocity vector and its horizontal and vertical components. **(1 pt)**



b. Find flight time, range, and velocity and acceleration at the top. **(3 pt)**

6. A ball leaves a 4 m ledge horizontally at 5 m/s. Use g=10 m/s² and consider airborne frames only. x^2 (m^2) -> y (m): (1.08, 3.784); (2.08, 3.584); (3.08, 3.384).

a. Build the model for airborne video frames from physics and predict both slope and intercept. **(2 pt)**

b. Label the axes and units and demonstrate interpolation between two readings. **(2 pt)**

7. A horizontal-launch experiment measures table height h and range R. Derive launch speed u and the speed needed for the same range at four times the height. **(3 pt)**